

Principles of Quantum Mechanics

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1 Hilbert Space

1.1 Definition

Definition. (Hilbert space). A Hilbert Space $\mathcal{H}$ is a vector space over $\mathbb{C}$ where

- (i) For $\phi, \psi \in \mathcal{H}$, $a\phi + b\psi = b\psi + a\phi \in \mathcal{H}$ for all $a, b \in \mathbb{C}$;
- (ii) $\phi + (\psi + \xi) = (\phi + \psi) + \xi$;
- (iii) $\exists 0 \in \mathcal{H}$ such that $\psi + 0 = \psi$ for all $\phi \in \mathcal{H}$.

We also have that $\mathcal{H}$ is equipped with an inner product $(\cdot, \cdot) : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ such that

- (i) $(\psi, \phi) = (\phi, \psi)$;
- (ii) $(\phi, a\psi) = a(\phi, \psi) \forall a \in \mathbb{C}$ but $(a\phi, \psi) = \bar{a}(\phi, \psi)$;
- (iii) $(\phi, \phi) \geq 0$ with equality if and only if $\phi = 0$.

The inner product gives a *norm* $\|\psi\| = \sqrt{(\psi, \psi)}$. This satisfies the Cauchy-Schwarz inequality

$$|(\phi, \psi)| \leq \|\psi\| \|\phi\|.$$

$\mathcal{H}$ is complete with this norm: a convergent sequence $\{\phi_n\}$ converges in $\mathcal{H}$, so we can take limits and derivatives.